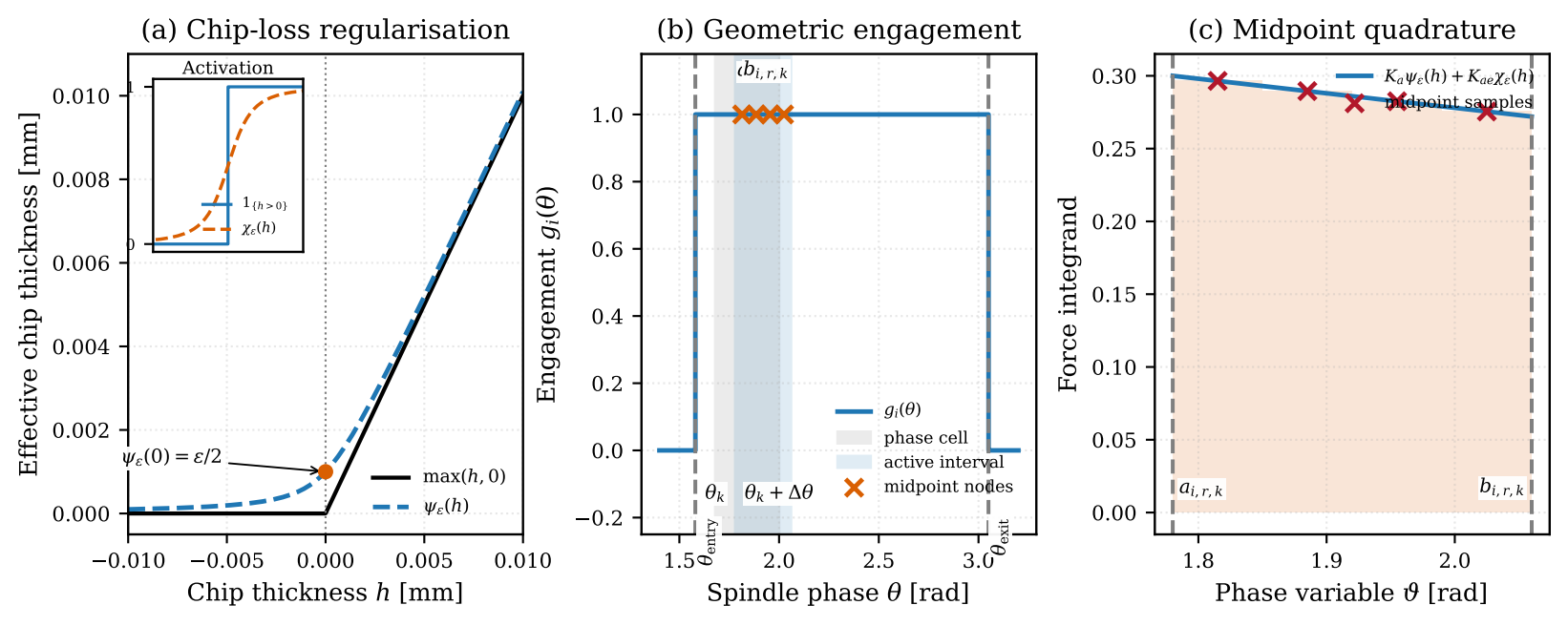
(a) Chip-loss regularisation
Activation
$1_{\{h>0\}}$
$\chi_\varepsilon(h)$
Effective chip thickness [mm]
Chip thickness h [mm]
$\psi_\varepsilon(0) = \varepsilon/2$
max(h, 0)
$\psi_\varepsilon(h)$
(b) Geometric engagement
Engagement $g_i(\theta)$
$b_{i,r,k}$
$g_i(\theta)$
phase cell
active interval
midpoint nodes
θ_k
$\theta_k + \Delta\theta$
θ_{entry}
θ_{exit}
Spindle phase θ [rad]
(c) Midpoint quadrature
Force integrand
$K_a\psi_\varepsilon(h) + K_{ae}\chi_\varepsilon(h)$
Midpoint samples
$a_{i,r,k}$
$b_{i,r,k}$
Phase variable ϑ [rad]